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Day 10

CSS

CSS Variables and Functions

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Variables

CSS Variables (aka custom properties) allow you to define reusable values throughout your CSS through root.

```
Syntax highlighting for CSS

:root {
 --primary-color: #3498db;
 --secondary-color: #2ecc71;
 --background-color: #ecf0f1;
 --text-color: #2c3e50;
 --font-family: 'Arial', sans-serif;
}
 body {
   font-family: var(--font-family);
   background-color: var(--background-color);
   color: var(--text-color);
   margin: 0;
   padding: 20px;
 }
```

Why Use CSS Variables?

Centralized styling
Easy theming
More readable and maintainable code
Dynamic styling with JavaScript
Responsive changes in media queries

Functions

CSS functions allow you to dynamically compute values for properties like
width, height, padding, font-size etc

It helps create responsive and flexible layout

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Common Functions

Function	Purpose / Use Case	Example
`calc()`	Math operations in CSS	`width: calc(100% - 2rem)`
`clamp()`	Responsive range (min, ideal, max)	`font-size: clamp(1rem, 2vw, 2rem)`
`min()`	Picks smallest value	`width: min(100%, 600px)`
`max()`	Picks largest value	`height: max(300px, 50vh)`
`var()`	Uses CSS variable	`color: var(--accent)`
`attr()`	Gets HTML attribute (only in content)	`content: attr(data-name)`
`env()`	System safe-area/environment values	`padding: env(safe-area-inset-top)`
`url()`	Load image, font, etc.	`background: url('bg.jpg')`
`rgb()` / `rgba()`	Define color (Red-Green-Blue)	`color: rgba(255, 0, 0, 0.5)`
`hsl()` / `hsla()`	Define color (Hue-Saturation-Lightness)	`color: hsl(200, 70%, 40%)`
`color-mix()`	Mixes two colors	`color: color-mix(in srgb, red, blue)`

Example

Balance: $8,200

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Code

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Balance Tag</title>
  <style>
    :root {
      --font-size: clamp(1rem, 2vw, 1.5rem);
      --padding: clamp(0.5rem, 1.5vw, 1rem);
      --radius: 100px;
      --low: #ef4444;
      --ok: #22c55e;
      --high: #3b82f6;
      --bg: #f1f5f9;
      --text: #0f172a;
      --duration: 0.4s ease;
    }

    body {
      margin: 0;
      height: 100vh;
      background: var(--bg);
      display: flex;
      align-items: center;
      justify-content: center;
      font-family: system-ui, sans-serif;
    }

    .balance-tag {
      font-size: var(--font-size);
      padding: var(--padding) calc(var(--padding) * 2);
      border-radius: var(--radius);
      color: white;
      background-color: var(--ok);
      transition: background-color var(--duration);
    }

    .balance-tag[data-state="low"] {
      background-color: var(--low);
    }

```

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Code

```
      .balance-tag {
        font-size: var(--font-size);
        padding: var(--padding) calc(var(--padding) * 2);
        border-radius: var(--radius);
        color: white;
        background-color: var(--ok);
        transition: background-color var(--duration);
      }

      .balance-tag[data-state="low"] {
        background-color: var(--low);
      }

      .balance-tag[data-state="high"] {
        background-color: var(--high);
      }
    </style>
  </head>
  <body>

    <div class="balance-tag" data-state="high">
      Balance: $8,200
    </div>

  </body>
  </html>

```